Myocardial Infarction

ORG: M-230 (ISC)
Link to Codes

MCG Health
Inpatient & Surgical Care
28th Edition

Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Admission to Inpatient Care

Note: For patients with clinically active secondary conditions, see Myocardial Infarction Multiple Condition Management Guidelines.

- Admission is indicated for **1 or more** of the following(1)(2)(3)(4)(5)(6)(7)(8):
  - Acute myocardial infarction (MI)[A][B] (not in context of cardiac procedure within last 48 hours), as indicated by **ALL** of the following(5)(13):
    - Elevated cardiac troponin level,[C][D][E][F] as indicated by **1 or more** of the following:
      - New or presumed new elevation of cardiac troponin level and emergent intervention (eg, invasive coronary angiography, cardiac surgery) indicated (eg, suspected STEMI, treatment refractory myocardial ischemia or hypotension, ventricular rupture)[G]
      - Initial troponin elevated with subsequent increase or decrease in level of 20% or more (ie, indicative of acute myocardial injury)
      - Decrease of cardiac troponin of 20% or more over sequential levels indicating acutely resolving myocardial injury (eg, myocardial injury more than 6 hours prior)
    - Myocardial injury due to acute ischemia, as indicated by **1 or more** of the following:
      - Symptoms consistent with myocardial ischemia (eg, chest pain, dyspnea)
      - New or presumed new ECG changes consistent with ischemia (eg, ST-segment change, left bundle branch block)
      - ECG with new or presumed new pathologic Q waves
      - Cardiac imaging with new or presumed new regional wall motion abnormality or loss of viable myocardium consistent with ischemia
      - Identification of a coronary thrombus (eg, by angiography, intracoronary imaging)
  - PCI-related myocardial infarction, as indicated by **ALL** of the following(5):
    - PCI procedure performed within last 48 hours
    - Preprocedure troponin level was not elevated, was stable (less than or equal to 20% change over serial testing), or was falling.
    - Postprocedure elevation of cardiac troponin, as indicated by **1 or more** of the following:

- Preprocedure cardiac troponin level not elevated with postprocedural cardiac troponin elevation to 5 times the upper limit of normal or higher[C][D][E][F][H]
- Preprocedure cardiac troponin level elevated with postprocedural cardiac troponin level that is **ALL** of the following:
  - Increased by 20% or more from preprocedural level
  - Elevated to 5 times the upper limit of normal or higher[H]

- Myocardial injury due to acute ischemia, as indicated by **1 or more** of the following:
  - New or presumed new ECG changes consistent with ischemia (eg, ST-segment change, left bundle branch block)
  - ECG with new or presumed new pathologic Q waves
  - Cardiac imaging with new or presumed new regional wall motion abnormality or loss of viable myocardium consistent with ischemia
  - Angiographic findings consistent with procedural flow limiting complication (eg, occlusion, embolization, dissection)

⊟ Cardiac procedure other than PCI-related (eg, transcatheter valve intervention, coronary artery bypass grafting) myocardial infarction, as indicated by **ALL** of the following(5):

- Cardiac procedure performed within last 48 hours
- Preprocedure troponin level was not elevated, was stable (less than or equal to 20% change over serial testing), or was falling.
- Postprocedure elevation of cardiac troponin, as indicated by **1 or more** of the following:
  - Preprocedure cardiac troponin level not elevated with postprocedural cardiac troponin elevation to 10 times the upper limit of normal or higher[C][D][E][F][H]
  - Preprocedure cardiac troponin level elevated with postprocedural cardiac troponin level that is **ALL** of the following:
    - Increased by 20% or more from preprocedural level
    - Elevated to 10 times the upper limit of normal or higher[H]
- Myocardial injury due to acute ischemia, as indicated by **1 or more** of the following:
  - ECG with new or presumed new pathologic Q waves
  - Cardiac imaging with new or presumed new regional wall motion abnormality or loss of viable myocardium consistent with ischemia
  - Angiographic findings of new graft occlusion or new native coronary artery occlusion

Alternatives to Admission

- Not usually used

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	• ICU[I] or intermediate care[J] after emergency treatment • Social Determinants of Health Assessment • Readmission Risk Assessment • Extended Stay Risk Assessment • Discharge planning	• **Clinical Indications met[K]**	• Bed rest • Possible commode privileges	• Parenteral and oral medications • Oral hydration • Clear liquid diet	• ECG, cardiac biomarkers • CXR • PT/PTT, platelet count[L] • PCI[M][N] • Oxygen[O]	• Possible IV morphine[P] • Anticoagulants • Antiplatelet agents (eg, aspirin, clopidogrel, ticagrelor, IV cangrelor) • Beta-blocker • Possible ACE inhibitor or ARB • Statin • Possible nitroglycerin
2	• Intermediate care or telemetry	• **Hemodynamic stability**	• Activity as tolerated	• **Oral hydration**	• **Oxygen absent**	• **IV nitroglycerin absent**

	• Social Determinants of Health Assessment • Readmission Risk Assessment • Extended Stay Risk Assessment	• **Chest pain, dyspnea, or anginal equivalent absent**	• Possibly begin ambulation	• **Oral medications** • Parenteral medications • Oral diet as tolerated	• ECG, cardiac biomarkers • PT/PTT and platelet count	• Anticoagulants • Antiplatelet agents (eg, aspirin, clopidogrel, ticagrelor) • Beta-blocker • Possible ACE inhibitor or ARB • Statin
3	• Social Determinants of Health Assessment • Readmission Risk Assessment • Extended Stay Risk Assessment • Intermediate care or telemetry to discharge • Complete discharge planning	• **Hemodynamic stability** • **Chest pain, dyspnea, or anginal equivalent absent** • **No evidence of bleeding** • **No evidence of recurrent myocardial ischemia** • **Dangerous arrhythmia absent** • **Vascular access site without evidence of infection, aneurysm, or growing hematoma** • **Renal function at baseline or acceptable for next level of care** • **Discharge plans and education understood**	• **Ambulatory or acceptable for next level of care**	• **Oral hydration[Q]** • **Oral medications or regimen acceptable for next level of care** • **Oral diet or acceptable for next level of care**	• Smoking cessation counseling if needed	• **Anticoagulants absent** • Antiplatelet agents (eg, aspirin, clopidogrel, ticagrelor) • Beta-blocker • Possible ACE inhibitor or ARB • Statin • Influenza vaccination[R]

(1)(2)(3)(4)(6)(8)(9)(10)(22)

Care Task List

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 2 days

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Note: See Myocardial Infarction Multiple Condition Management Benchmarking Table for more detailed information.

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(1)(3)(4)(6)(9)(10)(22):
  - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
    - Expect brief stay extension.
  - Hemodynamic instability, persisting symptoms after intensive medical management, or recurring severe prolonged symptoms(19)(23)

    - Anticipate inotrope, vasopressor, or mechanical circulatory support (eg, left ventricular assist device, extracorporeal membrane oxygenation) and possible early PCI of culprit lesion.
    - Anticipate staged revascularization of non-culprit lesion.
    - Expect brief to moderate stay extension.
    - See Hemodynamic Instability: Common Complications and Conditions ISC guideline as appropriate.
  - Intravascular procedural complications (eg, from percutaneous coronary intervention or placement of mechanical circulatory support device) such as acute vessel closure, stent thrombosis, stent malposition, pseudoaneurysm, arteriovenous fistula, atheroembolism, vessel dissection or perforation(24)(25)
    - Anticipate ultrasound or other imaging, topical compression, urgent repeat endovascular intervention, placement of arterial closure device, or surgery.
    - Expect minimal to brief stay extension for repeat intervention.
    - Expect brief to moderate stay extension for surgical intervention.
  - Extravascular procedural complications such as retroperitoneal hematoma, forearm hematoma and compartment syndrome, pericardial effusion, or cardiac tamponade(24)(25)
    - Anticipate imaging, echocardiography, pericardiocentesis, or surgery.
    - Expect brief stay extension.
  - Entry site complications causing bleeding, hematoma, or distal ischemia and requiring ongoing monitoring, surgical repair, or surgical thrombectomy(24)(25)
    - Expect minimal to brief stay extension.
  - Dangerous arrhythmia (26)
    - Urgent rhythm or rate control measures may be required.
    - Expect brief stay extension.
  - Complicated PCI (eg, unsuccessful PCI or PCI of non-native vessel)
    - Expect minimal to brief stay extension.
  - Mechanical complications of MI (eg, ventricular rupture, ventricular septal defect, valvular insufficiency, papillary muscle rupture, aneurysm or pseudoaneurysm)(27)(28)
    - Anticipate urgent imaging, possible mechanical support and PCI, and urgent surgical repair.
    - Expect moderate to prolonged stay extension.
  - Surgical revascularization via CABG(29)
    - Expect brief stay extension.
    - See Coronary Artery Bypass Graft (CABG) ISC guideline.
  - Heart failure (eg, pulmonary edema)(30)
    - Anticipate diuretic, afterload reduction, and inotropic intervention as indicated.
    - Pulmonary artery catheter, intra-aortic balloon pump, inotropic and vasopressor agents, ventricular assist device, noninvasive ventilation, or continuous positive airway pressure may be required for cardiogenic shock.(19)
    - Expect brief stay extension.
  - Postprocedural cerebrovascular accident(31)
    - Patients with underlying carotid artery disease, atrial fibrillation, and older age are at higher risk.
    - Anticipate brain imaging studies and neurologic monitoring.
    - Expect brief stay extension.
  - Postprocedural MI
    - Anticipate cardiac monitoring, MI-specific medication, serial ECGs, biomarker testing.
    - Expect brief stay extension.
  - Post-infarction pericarditis(32)
    - Anticipate cardiac imaging and treatment with NSAID, colchicine, and possible corticosteroid.
    - Expect brief stay extension.
  - Gastrointestinal bleed(33)
    - Patients who are older, in cardiogenic shock, or have a prior history of peptic ulcer, cirrhosis, anemia, or alcohol use disorder are at higher risk.
    - Anticipate endoscopic evaluation and treatment.
    - Expect brief stay extension.
  - Unstable pulmonary comorbidities (eg, COPD, pneumonia)(34)
    - Expect brief stay extension.
  - Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)
    - Expect monitoring, medication adjustment, and possible dialysis.
    - Expect brief stay extension.

See Common Complications and Conditions ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(1)(3)(4)(6)(8)(9)(12)(16)(18)(19)(22)(35):
  - Diagnostic test scheduling and completion, including:

        - Cardiac biomarkers, electrolytes, complete blood count, coagulation tests, hepatic and renal function tests
        - Toxicology screen (eg, cocaine, methamphetamine, marijuana)(7)
        - ECG
        - CXR
        - Echocardiogram(50)
        - Myocardial perfusion scan(50)
        - Cardiac MRI for MINOCA (myocardial infarction in the absence of coronary artery disease)(35)(50)
    - Treatment and procedure scheduling and completion, including(51):
        - Coronary angiography
        - PCI[M](2)(4)(52)
        - Fibrinolysis (if PCI not available)(2)
        - Mechanical circulatory support (eg, left ventricular assist device, extracorporeal membrane oxygenation)(19)(23)
        - Inotropes(23)(53)
        - Antiarrhythmics
        - Heparin or low-molecular-weight heparin(54)
        - Aspirin(54)
        - P2Y12 receptor inhibitor (eg, clopidogrel, prasugrel, ticagrelor, cangrelor)(54)
        - Glycoprotein IIb/IIIa receptor inhibitor
        - Thrombin inhibitor
        - Lipid-lowering therapy (eg, statin, cholesterol absorption inhibitor, monoclonal antibody)(6)(7)(55)
        - Beta-blocker(56)
        - Renin-angiotensin inhibitor (ACE inhibitor, angiotensin receptor blocker, or angiotensin receptor-neprilysin inhibitor (if diabetes or left ventricular dysfunction present))(7)
        - Aldosterone antagonist (if left ventricular dysfunction present)(7)
        - Morphine
        - Nitrates
        - Colchicine(57)(58)
        - Glycemic control(59)
        - Implantable cardiac defibrillator. See Electrophysiologic Study and Implantable Cardioverter-Defibrillator (ICD) Insertion ISC.
        - Temporary pacemaker
        - Transfusion
        - Aspirin allergy desensitization(60)
        - Influenza vaccination[R](21)
    - Consultation, assessment, and other services scheduling and completion, including:
        - Cardiology consultation
        - Cardiovascular surgery consultation
        - Cardiac rehabilitation(11)(16)(61)
        - Smoking cessation counseling and pharmacotherapy(62)
        - Nutrition counseling(62)
        - Palliative care consultation(16)
    - Identification of patient at high risk for readmission to prioritize transition and post-acute care:
        - Risk of readmission is increased by presence of **1 or more** of the following(44)(45)(48)(63)(64)(65)(66)(67):
            - Hospitalization (nonelective) in past 6 months(68)(69)(70)(71)
            - 2 or more emergency department visits in past 6 months(68)
            - No source of outpatient care other than emergency department (eg, no primary care provider)(71)(72)
            - Severe care transition barriers (eg, no caregiver, homeless)(69)(70)(73)
            - New-onset atrial fibrillation during hospitalization(47)
            - Age 80 years or older(46)
            - Symptomatic heart failure (past or present)(46)
            - Symptomatic COPD(74)
            - Anemia (hematocrit less than 30% (0.30))
            - Serum sodium less than 130 mEq/L (mmol/L)
            - Renal insufficiency (GFR less than 60 mL/min/1.73m^2 (1.00 mL/sec/1.73m^2))(69)(75) eGFR - Adult Calculator
            - Metastatic solid tumor (eg, lung cancer, breast cancer)
            - Advanced liver disease (eg, cirrhosis with portal hypertension, history of variceal bleed)
    - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool SR); key items include:
        - Ventricular arrhythmias
        - Cardiogenic shock
        - Refractory chest pain

- Bleeding
- Heart failure
- Acute renal failure
- Ventricular aneurysm or ventricular rupture
- Acute valvular regurgitation
- Depression(76)

Discharge

Discharge Planning

- Discharge planning includes[S]:
  - Assessment of needs and planning for care, including(78)(79):
    - Develop and modify treatment plan (involving multiple providers) as needed.
    - Evaluate and address preadmission functioning as needed.
    - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment SR for further information.(80)
    - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool SR for further information.(77)(81)
    - Evaluate and address patient or caregiver preferences as indicated.
    - Identify skilled services needed at next level of care, with specific attention to:
      - Cardiovascular status assessment(82)(83)
      - Medication management, adherence instruction, and side effects assessment(84)(85)
      - Psychosocial assessment, management, and referrals(86)(87)
      - Significant chronic condition exacerbation with need for clinical intervention and monitoring(82)
  - Early identification of anticipated discharge destination; options include(79)(88):
    - Home, considerations include:
      - Home safety assessment. See Home Safety Assessment SR for further information.
      - Patient safe to go home; examples include(89)(90)(91):
        - Medical status stable for patient's condition
        - Functional care can safely be provided with available resources.
        - Mental status stable for patient's condition
        - Medication availability confirmed and reconciliation complete
        - Patient/caregiver education completed with written discharge instructions provided
        - Community resources identified and referrals made, as needed
        - Home care arranged, if indicated
        - Necessary medical equipment delivery arranged or available in home, if indicated
        - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
      - Access to follow-up care
      - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment SR for further information.
      - Caregiver need, ability, and availability
    - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool SR for further information.
  - Transitions of care plan complete, including(79)(88)(92):
    - Patient and caregiver education complete. See Myocardial Infarction: Patient Education for Clinicians SR for further information.
      - See Teach Back Tool SR for further information.
    - Medication reconciliation completion includes(84)(93):
      - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
      - Assess each medication for correlation to disease state or medical condition.
      - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
      - Provide reconciled medication list to all treating providers.
      - Confirm that patient or caregiver can acquire medication.
      - Educate patient and caregiver.
        - Provide complete medication list to patient and caregiver.
        - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
        - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(94)

- Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
- See Medication Reconciliation Tool SR for further information.

- Plan communicated to patient, caregiver, and all members of care team, including(95):
  - Inpatient care and service providers
  - Primary care provider
  - All post-discharge care and service providers
- Appointments planned or scheduled, which may include:
  - Primary care provider
  - Cardiologist(96)
  - Cardiovascular surgeon
  - Dietitian
  - Follow-up evaluation call. See Post-Acute Care Follow-Up Call Tool SR for more information.
  - Rehabilitation therapy services(97)(98)(99)
  - Specialists for management of comorbidities as needed
  - Other
- Outpatient testing and procedure plans made, which may include:
  - Laboratory testing(100)
  - Other
- Referrals made for assistance or support, which may include:
  - Alcohol and other drug abuse or dependence treatment(86)(101)
  - Educational program (eg, chronic condition management, self-management)(102)(103)
  - Financial, for follow-up care, medication, and transportation
  - Self-help or support groups
  - Tobacco use treatment(104)(105)
  - Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
  - Blood pressure cuff
  - Other

Discharge Destination

- Post-hospital levels of admission may include:
  - Home.
  - Home healthcare. See Home Care Indications for Admission Section HC in Myocardial Infarction guideline in Home Care.
  - Recovery facility care. See Recovery Facility Care Indications for Admission Section RFC in Myocardial Infarction guideline in Recovery Facility Care.

Evidence Summary

Background

Acute coronary syndrome describes the clinical spectrum from unstable angina to NSTEMI to STEMI.(5)(6)(7)(8)(9) **(EG 2)** STEMI is characterized by symptoms of myocardial ischemia (eg, chest pain, dyspnea), ST-segment elevation on ECG, and an increased release of cardiac troponin from baseline.(5) **(EG 2)** NSTEMI also includes the presence of symptoms of myocardial ischemia and increased cardiac troponin release from baseline, but does not have ST-segment elevation on ECG.(5)(6)(7)(8)(9) **(EG 2)** The term non-ST-elevation acute coronary syndrome (NSTE-ACS) indicates an acute coronary syndrome without ST elevations and includes both NSTEMI and unstable angina.(5)(6)(7)(8)(9) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 6 published peer reviewed articles, 8 specialty society or other evidence-based guidelines, and 1 book section.

Analysis of an all-payer database shows a mean length of stay of 3.5 days for patients with a principal diagnosis of myocardial infarction admitted to inpatient care.(10) **(EG 3)** Specialty society guidelines and a narrative review state that inpatient admission is appropriate for patients with myocardial infarction, with some requiring specialized care in a coronary care unit.(3)(7)(8) **(EG 2)** A specialty guideline recommends that patients experiencing an acute STEMI should be triaged or transferred to a primary or comprehensive heart attack center that has 24/7 capability of performing PCI.(11) **(EG 2)** Some patients may be treated with thrombolysis prior to transfer if PCI cannot be performed within a 2-hour window.(2)(11) **(EG 2)**

Hospitalization

For STEMI, a specialty guideline recommends immediate (within 2 hours) primary PCI of culprit coronary lesions for patients whose symptom onset was within 24 hours of presentation, or for ongoing ischemia, heart failure, unstable rhythm disorder, or cardiogenic shock.(2) **(EG 2)** The same guideline recommends thrombolysis be used for STEMI only if PCI is not available within a 2-hour window, with subsequent transfer to a PCI-capable hospital for early angiography with intent to revascularize if needed.(2) **(EG 2)** A specialty guideline recommends that selected hemodynamically stable patients with STEMI treated initially with PCI of the identified culprit artery may undergo PCI of non-culprit arteries either as part of the initial procedure, as a second procedure before discharge, or as a planned interval procedure following discharge.(2) **(EG 2)** Systematic reviews with meta-analyses of randomized controlled trials have found that in patients with STEMI and multivessel disease, complete revascularization by PCI of non-infarct stenotic arteries resulted in reduced cardiac death and reduced incidence of reinfarction, while overall mortality reduction did not reach statistical significance.(36)(37)(38)(39) **(EG 1)** A subsequent randomized trial of 562 patients with acute myocardial infarction (47% STEMI, mean age 63 years) with non-culprit artery stenosis (greater than 50% stenosis) compared stenting of all stenoses (during index procedure or during index hospitalization) and stenting of only those with a measured fractional flow reserve (FFR) less than 0.80 and found that fewer FFR-guided patients had a PCI treated non-infarct artery (64.1% vs 97.1%, P less than 0.001) and that at a median of 3.5 years' follow-up, FFR-guided patients were less likely to experience the primary endpoint of death, myocardial infarction, repeat revascularization (7.4% vs 19.7%, hazard ratio (HR) 0.43 (95% confidence interval (CI) 0.25 to 0.75)).(40) **(EG 1)** In the same trial, FRR-guided patients had reduced all-cause mortality (2.1% vs 8.5%, HR 0.30 (95% CI 0.0.11 to 0.83)).(40) **(EG 1)** Another subsequent randomized trial including 1445 older patients with acute myocardial infarction (median age 80 years, 35% STEMI) compared non-culprit artery PCI (during index procedure or during index hospitalization) based on physiologic significant stenoses (eg, FFR, quantitative flow ratio) to no non-culprit artery PCI and found that 45% of identified non-culprit arteries (50% or more stenosis on angiography) in the treatment group were physiologically significant and were revascularized and that at 1-year follow-up, patients with complete revascularization had a lower rate of the composite primary outcome of death, myocardial infarction, stroke, or ischemia-driven revascularization (15.7% vs 21.0%, HR 0.73 (95% CI 0.57 to 0.93)) without increased safety outcomes (eg, bleeding, acute kidney injury).(41) **(EG 1)** In the same trial, physiologically driven complete revascularization also reduced all-cause death (9.2% vs 12.8%, HR 0.70 (95% CI 0.51 to 0.96)) and myocardial infarction (4.4% vs 7.0%, HR 0.62 (95% CI 0.40 to 0.97)).(41) **(EG 1)**

A systematic review and meta-analysis of 31 randomized controlled trials (27,071 patients) found that a radial rather than femoral approach to diagnostic cardiac catheterizations and PCI decreased the rate of access site complications and short-term (30 days) all-cause mortality.(42) **(EG 1)** A meta-analysis and trial sequence analysis of 17 randomized controlled trials (11,992 patients) comparing radial to femoral approach to PCI for STEMI found the radial approach had a lower 30-day mortality, fewer major adverse cardiac events (MACE) (this composite outcome varied by study; MACE classically includes nonfatal myocardial infarction, stroke, and cardiovascular mortality, but may also include all-cause mortality, heart failure hospitalization, and repeat revascularization procedure), less major bleeding, and fewer access site complications with no difference in postprocedure stroke or myocardial infarction.(43) **(EG 1)** A specialty society, based upon a systematic review of the randomized controlled trial literature, concludes that radial artery access is preferred for PCI, including patients with acute coronary syndromes, due to reduced rates of major bleeding, vascular complications, and mortality (in acute coronary syndrome (ACS) patients).(2) **(EG 2)**

Readmission risk and reduction: An administrative-claims-based and medical-record-based analysis of over 200,000 Medicare patients (mean age 79 years) admitted for an acute MI at one of 4171 hospitals found that risk factors associated with an increased likelihood of 30-day all-cause hospital readmission included patient history of heart failure, renal insufficiency, and COPD.(44) **(EG 2)** The same analysis found that additional risk factors for readmission based upon clinical findings present during the index acute MI hospitalization included evidence of heart failure (eg, pulmonary congestion on chest x-ray, rales on examination), anemia (hematocrit less than 30% (0.30)), and a serum sodium of less than 130 mEq/L (mmol/L).(44) **(EG 2)** Analysis of 5571 patients hospitalized for ST-elevation MI found, after multivariate adjustment, that comorbid COPD was an independent risk factor for nonelective 30-day hospital readmission.(45) **(EG 2)** Analysis of a prospective database including 1271 patients hospitalized with acute ST-elevation MI found, after multivariate analysis, that age 80 years or older and new-onset heart failure were independent risk factors for emergent 30-day readmission or death.(46) **(EG 2)** Multivariate analysis of a cohort including 6384 patients with acute myocardial infarction found that new-onset atrial fibrillation during hospitalization was an independent risk factor for readmission at 30 days.(47) **(EG 2)** Logistic regression analysis of 212,171 patients discharged after hospitalization for acute myocardial infarction found that end-stage renal disease was independently associated with an increased 30-day readmission rate.(48) **(EG 2)** Multivariate analysis of 43,205 patients (mean age 64 years) with acute ST-elevation MI reported a 12.8% 30-day readmission rate and found that a Charlson comorbidity score of 5 or greater was an independent risk factor for 30-day readmission.(49) **(EG 2)**

Length of Stay

Analysis of national hospital discharge data shows 50% of hospitalized patients with the principal diagnosis of NSTEMI discharged in 2 days or less.(10) **(EG 3)** Analysis of national hospital discharge data shows 58% of hospitalized patients with the principal diagnosis of STEMI discharged in 2 days or less.(10) **(EG 3)**

Rationale

Use of this MCG care guideline helps the clinician identify patient-specific complex clinical factors that make it reasonable to expect a necessity for hospital care across 2 or more midnights. The evidence-based clinical indication criteria assist the clinician in the decision to appropriately admit a patient to inpatient care. For Medicare enrollees, MCG care guidelines also support the clinician in the decision to appropriately admit a patient to inpatient care when there is not an expectation of 2 or more midnights of hospital care (eg, CMS' Two-Midnight Rule exceptions).

Use of these evidence-based clinical criteria to support decision making around the need for inpatient treatment is of clinical benefit to the patient and is crucial for quality patient care. Use of evidence-based criteria ensures patients receive the most appropriate care for their specific condition, reduces unnecessary risks, promotes recovery, and provides a standard that can reduce disparities in care. Not only do evidence-based criteria help align treatment decisions with best practice, but they can also help reduce unwarranted variation in care by fostering consistency and equality in healthcare provision across regions and facilities. Appropriate use of the evidence-based clinical criteria in conjunction with the Supplemental Medicare Criteria ensures Medicare patients receive full access to Medicare benefits (eg, inpatient care), without risk of delay or decreased access, based on variables such as clinical severity of illness, length of provision of medically necessary hospital-level care, or satisfaction of specified Medicare criteria as directed by CMS. Use of these criteria will help ensure that the patient receives necessary care in the appropriate setting.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(17); 42 CFR 419.22(106); 42 CFR 422.101(107)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(108); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(109); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(110); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(111)

Medicare Coverage Determinations: Medicare Coverage Database(112)

References

1. O'Gara PT, et al. 2013 ACCF/AHA Guideline for the management of ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. Circulation 2013;127(4):e362-e425. DOI: 10.1161/CIR.0b013e3182742cf6. (Reaffirmed 2022 Jun) [Context Link 1, 2, 3, 4, 5, 6, 7] View abstract...
2. Lawton JS, et al. 2021 ACC/AHA/SCAI guideline for coronary artery revascularization: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 2022;145(3):e18-e114. DOI: 10.1161/CIR.0000000000001038. (Reaffirmed 2023 Aug) [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16] View abstract...
3. Amsterdam EA, et al. AHA/ACC guideline for the management of patients with non-ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. Circulation 2014;130(25):e344-e426. DOI: 10.1161/CIR.0000000000000134. (Reaffirmed 2023 Jul) [Context Link 1, 2, 3, 4, 5, 6] View abstract...
4. Patel MR, et al. ACC/AATS/AHA/ASE/ASNC/SCAI/SCCT/STS 2016 appropriate use criteria for coronary revascularization in patients with acute coronary syndromes: a report of the American College of Cardiology appropriate use criteria task force, American Association for Thoracic Surgery, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Society for Cardiovascular Angiography and Interventions, Society of Cardiovascular Computed Tomography, and the Society of Thoracic Surgeons. Journal of the American College of Cardiology 2017;69(5):570-591. DOI: 10.1016/j.jacc.2016.10.034. (Reaffirmed 2023 Jun) [Context Link 1, 2, 3, 4, 5] View abstract...
5. Thygesen K, et al. Fourth Universal Definition of Myocardial Infarction (2018). Circulation 2018;138(20):e618-e651. DOI: 10.1161/CIR.0000000000000617. [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32] View abstract...
6. Bergmark BA, Mathenge N, Merlini PA, Lawrence-Wright MB, Giugliano RP. Acute coronary syndromes. Lancet 2022;399(10332):1347-1358. DOI: 10.1016/S0140-6736(21)02391-6. [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11] View abstract...
7. Bhatt DL, Lopes RD, Harrington RA. Diagnosis and treatment of acute coronary syndromes: a review. Journal of the American Medical Association 2022;327(7):662-675. DOI: 10.1001/jama.2022.0358. [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13] View abstract...
8. Collet JP, et al. 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. European Heart Journal 2021;42(14):1289-1367. DOI: 10.1093/eurheartj/ehaa575. (Reaffirmed 2022 Jul) [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13] View abstract...
9. Giugliano RP, Braunwald E. Non-ST elevation acute coronary syndromes. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:714-738. [Context Link 1, 2, 3, 4, 5, 6]
10. Premier PINC AI™ Healthcare Database (PHD), 01/01/2021-12/31/2022. Premier, Inc. [Context Link 1, 2, 3, 4, 5]
11. Jacobs AK, et al. Systems of care for ST-segment-elevation myocardial infarction: a policy statement from the American Heart Association. Circulation 2021;144(20):e310-e327. DOI: 10.1161/CIR.0000000000001025. (Reaffirmed 2021 Nov) [Context Link 1, 2, 3, 4] View abstract...
12. Sandoval Y, Jaffe AS. Type 2 myocardial infarction: JACC review topic of the week. Journal of the American College of Cardiology 2019;73(14):1846-1860. DOI: 10.1016/j.jacc.2019.02.018. [Context Link 1, 2] View abstract...
13. Twerenbold R, et al. Clinical use of high-sensitivity cardiac troponin in patients with suspected myocardial infarction. Journal of the American College of Cardiology 2017;70(8):996-1012. DOI: 10.1016/j.jacc.2017.07.718. [Context Link 1] View abstract...
14. Gulati M, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 2021;144(22):e368-e454. DOI: 10.1161/CIR.0000000000001029. (Reaffirmed 2023 Jul) [Context Link 1] View abstract...

15. Azar RR, Sarkis A, Giannitsis E. A practical approach for the use of high-sensitivity cardiac troponin assays in the evaluation of patients with chest pain. American Journal of Cardiology 2021;139:1-7. DOI: 10.1016/j.amjcard.2020.10.037. [Context Link 1, 2] View abstract...
16. Damluji AA, et al. Management of acute coronary syndrome in the older adult population: a scientific statement from the American Heart Association. Circulation 2023;147(3):e32-e62. DOI: 10.1161/CIR.0000000000001112. (Reaffirmed 2023 Mar 07) [Context Link 1, 2, 3, 4, 5] View abstract...
17. Centers for Medicare and Medicaid Services. "Admissions." 42 CFR 412.3 Washington, DC 2023 Jul [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-412. [Context Link 1, 2]
18. van Diepen S, et al. Contemporary management of cardiogenic shock: a scientific statement from the American Heart Association. Circulation 2017;136(16):e232-e268. DOI: 10.1161/CIR.0000000000000525. [Context Link 1, 2] View abstract...
19. Henry TD, et al. Invasive management of acute myocardial infarction complicated by cardiogenic shock: a scientific statement from the American Heart Association. Circulation 2021;143(15):e815-e829. DOI: 10.1161/CIR.0000000000000959. (Reaffirmed 2023 Jul) [Context Link 1, 2, 3, 4, 5] View abstract...
20. Chioncel O, et al. Epidemiology, pathophysiology and contemporary management of cardiogenic shock - a position statement from the Heart Failure Association of the European Society of Cardiology. European Journal of Heart Failure 2020;22(8):1315-1341. DOI: 10.1002/ejhf.1922. [Context Link 1] View abstract...
21. Maniar YM, Al-Abdouh A, Michos ED. Influenza vaccination for cardiovascular prevention: further insights from the IAMI trial and an updated meta-analysis. Current Cardiology Reports 2022;24(10):1327-1335. DOI: 10.1007/s11886-022-01748-8. [Context Link 1, 2] View abstract...
22. McDaniel M. Acute coronary syndromes. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:929-940. [Context Link 1, 2, 3]
23. Samsky MD, Morrow DA, Proudfoot AG, Hochman JS, Thiele H, Rao SV. Cardiogenic shock after acute myocardial infarction: a review. Journal of the American Medical Association 2021;326(18):1840-1850. DOI: 10.1001/jama.2021.18323. [Context Link 1, 2, 3] View abstract...
24. Sandoval Y, Bell MR, Gulati R. Transradial artery access complications. Circulation. Cardiovascular Interventions 2019;12(11):e007386. DOI: 10.1161/CIRCINTERVENTIONS.119.007386. [Context Link 1, 2, 3] View abstract...
25. Damluji AA, et al. Position statement on vascular access safety for percutaneous devices in AMI complicated by cardiogenic shock. JACC. Cardiovascular Interventions 2022;15(20):2003-2019. DOI: 10.1016/j.jcin.2022.08.041. [Context Link 1, 2, 3] View abstract...
26. Vallabhajosyula S, et al. Burden of arrhythmias in acute myocardial infarction complicated by cardiogenic shock. American Journal of Cardiology 2020;125(12):1774-1781. DOI: 10.1016/j.amjcard.2020.03.015. [Context Link 1] View abstract...
27. Damluji AA, et al. Mechanical complications of acute myocardial infarction: a scientific statement from the American Heart Association. Circulation 2021;144(2):Online. DOI: 10.1161/CIR.0000000000000985. [Context Link 1] View abstract...
28. Gong FF, Vaitenas I, Malaisrie SC, Maganti K. Mechanical complications of acute myocardial infarction: a review. Journal of the American Medical Association Cardiology 2021;6(3):341-349. DOI: 10.1001/jamacardio.2020.3690. [Context Link 1] View abstract...
29. Sugiyama T, Hasegawa K, Kobayashi Y, Takahashi O, Fukui T, Tsugawa Y. Differential time trends of outcomes and costs of care for acute myocardial infarction hospitalizations by ST elevation and type of intervention in the United States, 2001-2011. Journal of the American Heart Association 2015;4(3):e001445. DOI: 10.1161/JAHA.114.001445. [Context Link 1] View abstract...
30. Pasala S, et al. The influence of heart failure on clinical and economic outcomes among older adults =75 years of age with acute myocardial infarction. American Heart Journal 2022;246:65-73. DOI: 10.1016/j.ahj.2021.11.021. [Context Link 1] View abstract...
31. Alkhouli M, Alqahtani F, Tarabishy A, Sandhu G, Rihal CS. Incidence, predictors, and outcomes of acute ischemic stroke following percutaneous coronary intervention. JACC. Cardiovascular Interventions 2019;12(15):1497-1506. DOI: 10.1016/j.jcin.2019.04.015. [Context Link 1] View abstract...
32. Verma BR, et al. Pericarditis and post-cardiac injury syndrome as a sequelae of acute myocardial infarction. Current Cardiology Reports 2020;22(10):Online. DOI: 10.1007/s11886-020-01371-5. [Context Link 1] View abstract...
33. Albeiruti R, Chaudhary F, Alqahtani F, Kupec J, Balla S, Alkhouli M. Incidence, predictors, and outcomes of gastrointestinal bleeding in patients admitted with ST-elevation myocardial infarction. American Journal of Cardiology 2019;124(3):343-348. DOI: 10.1016/j.amjcard.2019.05.008. [Context Link 1] View abstract...
34. Vallabhajosyula S, Patlolla SH, Murphree DH, Cheungpasitporn W, Holme DR, Gersh BJ. Temporal trends, management and outcomes of acute myocardial infarction with concomitant respiratory infections. American Journal of Cardiology 2021;150:1-7. DOI: 10.1016/j.amjcard.2021.03.037. [Context Link 1] View abstract...
35. Tamis-Holland JE, et al. Contemporary diagnosis and management of patients with myocardial infarction in the absence of obstructive coronary artery disease: a scientific statement from the American Heart Association. Circulation 2019;139(18):e891-e908. DOI: 10.1161/CIR.0000000000000670. [Context Link 1, 2] View abstract...
36. Ahmad Y, et al. Complete revascularization by percutaneous coronary intervention for patients with ST-segment-elevation myocardial nfarction and multivessel coronary artery disease: an updated meta-analysis of randomized trials. Journal of the American Heart Association 2020;9(12):Online. DOI: 10.1161/JAHA.119.015263. [Context Link 1] View abstract...
37. Pavasini R, et al. Complete revascularization reduces cardiovascular death in patients with ST-segment elevation myocardial infarction and multivessel disease: systematic review and meta-analysis of randomized clinical trials. European Heart Journal 2020;41(42):4103-4110. DOI: 10.1093/eurheartj/ehz896. [Context Link 1] View abstract...
38. Bainey KR, et al. Complete vs culprit-lesion-only revascularization for ST-segment elevation myocardial infarction: a systematic review and meta-analysis. Journal of the American Medical Association Cardiology 2020;5(8):1-9. DOI: 10.1001/jamacardio.2020.1251. [Context Link 1] View abstract...

39. Atti V, et al. Multivessel versus culprit-only revascularization in STEMI and multivessel coronary artery disease: meta-analysis of randomized trials. JACC. Cardiovascular Interventions 2020;13(13):1571-1582. DOI: 10.1016/j.jcin.2020.04.055. [Context Link 1] View abstract...
40. Lee JM, et al. Fractional flow reserve versus angiography-guided strategy in acute myocardial infarction with multivessel disease: a randomized trial. European Heart Journal 2023;44(6):473-484. DOI: 10.1093/eurheartj/ehac763. [Context Link 1, 2] View abstract...
41. Biscaglia S, et al. Complete or culprit-only PCI in plder patients with myocardial infarction. New England Journal of Medicine 2023;389(10):889-898. DOI: 10.1056/NEJMoa2300468. [Context Link 1, 2] View abstract...
42. Kolkailah AA, Alreshq RS, Muhammed AM, Zahran ME, Anas El-Wegoud M, Nabhan AF. Transradial versus transfemoral approach for diagnostic coronary angiography and percutaneous coronary intervention in people with coronary artery disease. Cochrane Database of Systematic Reviews 2018, Issue 4. Art. No.: CD012318. DOI: 10.1002/14651858.CD012318.pub2. [Context Link 1] View abstract...
43. Osman M, et al. Radial versus femoral access for percutaneous coronary intervention in patients with ST-segment elevation myocardial infarction: Trial sequential analysis. American Heart Journal 2020;224:98-104. DOI: 10.1016/j.ahj.2020.03.014. [Context Link 1] View abstract...
44. Krumholz HM, et al. An administrative claims measure suitable for profiling hospital performance based on 30-day all-cause readmission rates among patients with acute myocardial infarction. Circulation. Cardiovascular Quality and Outcomes 2011;4(2):243-252. DOI: 10.1161/CIRCOUTCOMES.110.957498. [Context Link 1, 2, 3] View abstract...
45. Kociol RD, et al. International variation in and factors associated with hospital readmission after myocardial infarction. Journal of the American Medical Association 2012;307(1):66-74. DOI: 10.1001/jama.2011.1926. [Context Link 1, 2] View abstract...
46. Brown JR, Conley SM, Niles NW. Predicting readmission or death after acute ST-elevation myocardial infarction. Clinical Cardiology 2013;36(10):570-575. DOI: 10.1002/clc.22156. [Context Link 1, 2, 3] View abstract...
47. Kundu A, et al. Relation of atrial fibrillation in acute myocardial infarction to in-hospital complications and early hospital readmission. American Journal of Cardiology 2016;117(8):1213-1218. DOI: 10.1016/j.amjcard.2016.01.012. [Context Link 1, 2] View abstract...
48. Khera R, et al. Comparison of readmission rates after acute myocardial infarction in 3 patient age groups (18 to 44, 45 to 64, and =65 Years) in the United States. American Journal of Cardiology 2017;120(10):1761-1767. DOI: 10.1016/j.amjcard.2017.07.081. [Context Link 1, 2] View abstract...
49. Sud K, et al. Readmissions in ST-elevation myocardial infarction and cardiogenic shock (from Nationwide Readmission Database). American Journal of Cardiology 2019;124(12):1841-1850. DOI: 10.1016/j.amjcard.2019.08.048. [Context Link 1] View abstract...
50. Edvardsen T, et al. Non-invasive imaging in Coronary syndromes: recommendations of the European Association of Cardiovascular Imaging and the American Society of Echocardiography, in Collaboration with the American Society of Nuclear Cardiology, Society of Cardiovascular Computed Tomography, and Society for Cardiovascular Magnetic Resonance. Journal of the American Society of Echocardiography 2022;35(4):329-354. DOI: 10.1016/j.echo.2021.12.012. (Reaffirmed 2022 Jul) [Context Link 1, 2, 3] View abstract...
51. Szummer K, Jernberg T, Wallentin L. From early pharmacology to recent pharmacology interventions in acute coronary syndromes: JACC state-of-the-art review. Journal of the American College of Cardiology 2019;74(12):1618-1636. DOI: 10.1016/j.jacc.2019.03.531. [Context Link 1] View abstract...
52. Diletti R, et al. Immediate versus staged complete revascularisation in patients presenting with acute coronary syndrome and multivessel coronary disease (BIOVASC): a prospective, open-label, non-inferiority, randomised trial. Lancet 2023;401(10383):1172-1182. DOI: 10.1016/S0140-6736(23)00351-3. [Context Link 1] View abstract...
53. Uhlig K, et al. Inotropic agents and vasodilator strategies for the treatment of cardiogenic shock or low cardiac output syndrome. Cochrane Database of Systematic Reviews 2020, Issue 11. Art. No.: CD009669. DOI: 10.1002/14651858.CD009669.pub4. [Context Link 1] View abstract...
54. Rodriguez F, Harrington RA. Management of antithrombotic therapy after acute coronary syndromes. New England Journal of Medicine 2021;384(5):452-460. DOI: 10.1056/NEJMra1607714. [Context Link 1, 2, 3] View abstract...
55. Newman CB, et al. Statin safety and associated adverse events: a scientific statement from the American Heart Association. Arteriosclerosis, Thrombosis, and Vascular Biology 2019;39(2):e38-e81. DOI: 10.1161/ATV.0000000000000073. (Reaffirmed 2023 Apr) [Context Link 1] View abstract...
56. Safi S, Sethi NJ, Nielsen EE, Feinberg J, Jakobsen JC, Gluud C. Beta-blockers for suspected or diagnosed acute myocardial infarction. Cochrane Database of Systematic Reviews 2019, Issue 12. Art. No.: CD012484. DOI: 10.1002/14651858.CD012484.pub2. [Context Link 1] View abstract...
57. Tardif JC, et al. Efficacy and safety of low-dose colchicine after myocardial infarction. New England Journal of Medicine 2019;381(26):2497-2505. DOI: 10.1056/NEJMoa1912388. [Context Link 1] View abstract...
58. Bouabdallaoui N, et al. Time-to-treatment initiation of colchicine and cardiovascular outcomes after myocardial infarction in the Colchicine Cardiovascular Outcomes Trial (COLCOT). European Heart Journal 2020;41(42):4092-4099. DOI: 10.1093/eurheartj/ehaa659. [Context Link 1] View abstract...
59. American Diabetes Association. Standards of medical care in diabetes - 2023. Diabetes Care 2023;45(Supplement 1):S1-S291. [Context Link 1]
60. Cook KA, White AA. Rapid aspirin challenge in patients with aspirin allergy and acute coronary syndromes. Current Allergy and Asthma Reports 2016;16(2):11. DOI: 10.1007/s11882-015-0593-2. [Context Link 1] View abstract...
61. Dibben G, et al. Exercise-based cardiac rehabilitation for coronary heart disease. Cochrane Database of Systematic Reviews 2021, Issue 11. Art. No.: CD001800. DOI: 10.1002/14651858.CD001800.pub4. [Context Link 1] View abstract...
62. Writing Committee Members, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA guideline for the management of patients with chronic coronary disease: a report of the American Heart Association/American College of Cardiology joint committee on clinical practice guidelines. Journal of the American College of Cardiology 2023;82(9):833-955. DOI: 10.1016/j.jacc.2023.04.003. [Context Link 1, 2] View abstract...
63. MarketScan Database, 2012-2013 (Copyright @2012-2013 Truven Health Analytics Inc. All Rights Reserved.); proprietary health insurance data sources (2013-2014); and Medicare 100% Standard Analytical File (2013). [Context Link 1]

64. Preyde M, Brassard K. Evidence-based risk factors for adverse health outcomes in older patients after discharge home and assessment tools: a systematic review. Journal of Evidence-Based Social Work 2011;8(5):445-68. DOI: 10.1080/15433714.2011.542330. [Context Link 1] View abstract...

65. Billings J, Dixon J, Mijanovich T, Wennberg D. Case finding for patients at risk of readmission to hospital: development of algorithm to identify high risk patients. British Medical Journal 2006;333(7563):327. DOI: 10.1136/bmj.38870.657917.AE. [Context Link 1] View abstract...

66. Hasan O, et al. Hospital readmission in general medicine patients: a prediction model. Journal of General Internal Medicine 2010;25(3):211-9. DOI: 10.1007/s11606-009-1196-1. [Context Link 1] View abstract...

67. Billings J, Blunt I, Steventon A, Georghiou T, Lewis G, Bardsley M. Development of a predictive model to identify inpatients at risk of re-admission within 30 days of discharge (PARR-30). BMJ Open 2012;2(4):e001667. DOI: 10.1136/bmjopen-2012-001667. [Context Link 1] View abstract...

68. van Walraven C, et al. Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community. Canadian Medical Association Journal 2010;182(6):551-7. DOI: 10.1503/cmaj.091117. [Context Link 1, 2] View abstract...

69. Jencks SF, Williams MV, Coleman EA. Rehospitalizations among patients in the Medicare fee-for-service program. New England Journal of Medicine 2009;360(14):1418-28. DOI: 10.1056/NEJMsa0803563. [Context Link 1, 2, 3] View abstract...

70. Garcia-Perez L, Linertova R, Lorenzo-Riera A, Vazquez-Diaz JR, Duque-Gonzalez B, Sarria-Santamera A. Risk factors for hospital readmissions in elderly patients: a systematic review. Quarterly Journal of Medicine 2011;104(8):639-51. DOI: 10.1093/qjmed/hcr070. [Context Link 1, 2] View abstract...

71. Woz S, et al. Gender as risk factor for 30 days post-discharge hospital utilisation: a secondary data analysis. BMJ Open 2012;2(2):e000428. DOI: 10.1136/bmjopen-2011-000428. [Context Link 1, 2] View abstract...

72. Jack BW, et al. A reengineered hospital discharge program to decrease rehospitalization: a randomized trial. Annals of Internal Medicine 2009;150(3):178-87. [Context Link 1] View abstract...

73. Arbaje AI, Wolff JL, Yu Q, Powe NR, Anderson GF, Boult C. Postdischarge environmental and socioeconomic factors and the likelihood of early hospital readmission among community-dwelling Medicare beneficiaries. Gerontologist 2008;48(4):495-504. [Context Link 1] View abstract...

74. Campo G, et al. Impact of COPD on long-term outcome after ST-segment elevation myocardial infarction receiving primary percutaneous coronary intervention. Chest 2013;144(3):750-7. DOI: 10.1378/chest.12-2313. [Context Link 1] View abstract...

75. Silverstein MD, Qin H, Mercer SQ, Fong J, Haydar Z. Risk factors for 30-day hospital readmission in patients =65 years of age. Proceedings (Baylor University. Medical Center) 2008;21(4):363-72. [Context Link 1] View abstract...

76. Lichtman JH, et al. Depression as a risk factor for poor prognosis among patients with acute coronary syndrome: systematic review and recommendations: a scientific statement from the American Heart Association. Circulation 2014;129(12):1350-1369. DOI: 10.1161/CIR.0000000000000019. [Context Link 1] View abstract...

77. Hudson T. The role of social determinates of health in discharge practices. Nursing Clinics of North America 2021;56(3):369-378. DOI: 10.1016/j.cnur.2021.04.004. [Context Link 1, 2] View abstract...

78. Mayer RS, Noles A, Vinh D. Determination of postacute hospitalization level of care. Medical Clinics of North America 2020;104(2):345-357. DOI: 10.1016/j.mcna.2019.10.011. [Context Link 1] View abstract...

79. Centers for Medicare and Medicaid Services. "Condition of participation: Discharge planning." 42 CFR Pt. 482.43. Washington, DC 2022 Oct [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/. [Context Link 1, 2, 3]

80. Gulati R, et al. Acute myocardial infarction in young individuals. Mayo Clinic Proceedings 2020;95(1):136-156. DOI: 10.1016/j.mayocp.2019.05.001. [Context Link 1] View abstract...

81. Balla S, Alqahtani F, Alhajji M, Alkhouli M. Cardiovascular outcomes and rehospitalization rates in homeless patients admitted with acute myocardial infarction. Mayo Clinic Proceedings 2020;95(4):660-668. DOI: 10.1016/j.mayocp.2020.01.013. [Context Link 1] View abstract...

82. The client with alterations in cardiovascular function. In: Haugen N, Galura SJ, editors. Ulrich & Canale's Nursing Care Planning Guides. 8th ed. St. Louis, MO: Elsevier; 2020:217-339. [Context Link 1, 2]

83. Hagler D, Hagler DR. Assessment: cardiovascular system. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:767-796. [Context Link 1]

84. Adverse drug reactions and medication errors. In: Burchum JR, Rosenthal LD, editors. Lehne's Pharmacology for Nursing Care. 11th ed. Elsevier; 2022:63-73. [Context Link 1, 2]

85. Kumar K, Golwala H. Antiplatelet agents in acute ST elevation myocardial infarction. American Journal of Medicine 2022;135(6):697-708. DOI: 10.1016/j.amjmed.2022.01.042. [Context Link 1] View abstract...

86. Mora S, Libby P, Ridker PM. Primary prevention of cardiovascular disease. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:442-470. [Context Link 1, 2]

87. Freedland KE, Carney RM, Lenze EJ, Rich MW. Psychiatric and psychosocial aspects of cardiovascular disease. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1841-1852. [Context Link 1]

88. Saleeby J. Communication and collaboration. In: Perry AG, Potter PA, Ostendorf WR, editors. Nursing Interventions and Clinical Skills. 7th ed. Elsevier; 2020:9-21. [Context Link 1, 2]

89. Elements of Excellence in Transitions of Care (TOC). TOC Checklist [Internet] National Transitions of Care Coalition. Accessed at: https://www.ntocc.org. [accessed 2023 Aug 10] [Context Link 1]

90. Medical-surgical nursing. In: Hinkle JL, Cheever KH, Overbaugh KJ, editors. Brunner & Suddarth's Textbook of Medical-Surgical Nursing. 15th ed. Wolters Kluwer; 2022:33-55. [Context Link 1]

91. Transitions of care. In: Gillingham DC, editor. Foundations of Case Management. Blue Bayou Press; 2021:137-144. [Context Link 1]

92. Becker C, et al. Interventions to improve communication at hospital discharge and rates of readmission: a systematic review and meta-analysis. JAMA Network Open 2021;4(8):e2119346. DOI: 10.1001/jamanetworkopen.2021.19346. [Context Link 1] View abstract...
93. The nursing process in drug therapy and patient safety. In: Tucker RG, editor. Karch's Focus on Nursing Pharmacology. 9th ed. Wolters Kluwer; 2023:47-57. [Context Link 1]
94. Ostendorf WR. Preparation for safe medication administration. In: Perry AG, Potter PA, Ostendorf WR, editors. Nursing Interventions and Clinical Skills. 7th ed. Elsevier; 2020:551-567. [Context Link 1]
95. The case manager's roles, functions, and activities. In: Gillingham DC, editor. Foundations of Case Management. Blue Bayou Press; 2021:9-14. [Context Link 1]
96. Bohula EA, Morrow DA. ST-elevation myocardial infarction: management. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:662-713. [Context Link 1]
97. Cardiac Rehabilitation. [Internet] National Heart Lung and Blood Institute. Accessed at: https://www.nhlbi.nih.gov/health-topics/cardiac-rehabilitation. Updated 2022 Mar [accessed 2023 Oct 23] [Context Link 1]
98. Thomas RJ. Comprehensive cardiac rehabilitation. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:588-592. [Context Link 1]
99. McBride JM, Thananopavarn P, Tilley VST. Rehabilitation. In: Warshaw GA, Potter JF, Flaherty E, Heflin MT, McNabney MK, Ham RJ, editors. Ham's Primary Care Geriatrics. 7th ed. Elsevier; 2022:133-146. [Context Link 1]
100. Scirica BM, Libby P, Morrow DA. ST-elevation myocardial infarction: pathophysiology and clinical evolution. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:636-661. [Context Link 1]
101. Harding MM. Substance use disorders in acute care. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:162-178. [Context Link 1]
102. Giannopoulos G, et al. Usefulness of a structured adult education program in modifying markers of cardiovascular risk after acute myocardial infarction. American Journal of Cardiology 2020;125(6):845-850. DOI: 10.1016/j.amjcard.2019.12.033. [Context Link 1] View abstract...
103. Nursing diagnoses in alphabetical order. In: Doenges ME, Moorhouse MF, Murr AC, editors. Nursing Diagnosis Manual: Planning, Individualizing, and Documenting Client Care. 7th ed. F.A. Davis Company; 2022:40-1095. [Context Link 1]
104. Wu AD, et al. Smoking cessation for secondary prevention of cardiovascular disease. Cochrane Database of Systematic Reviews 2022, Issue 8. Art. No.: CD014936. DOI: 10.1002/14651858.CD014936.pub2. [Context Link 1] View abstract...
105. Mir H, Pipe A. Smoking cessation. In: Levine GN, editor. Cardiology Secrets. 6th ed. Elsevier; 2023:377-381. [Context Link 1]
106. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/. [Context Link 1]
107. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2023 Jun 05 [accessed 2023 Oct 30] Accessed at: https://www.ecfr.gov/. [Context Link 1]
108. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 1 - Inpatient Hospital Services Covered Under Part A Rev. 10892 [Internet] Centers for Medicare and Medicaid Services. 2021 Aug Accessed at: https://www.cms.gov/manuals/. [accessed 2023 Sep 08] [Context Link 1]
109. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 6 - Hospital Services Covered Under Part B Rev. 10541 [Internet] Centers for Medicare and Medicaid Services. 2020 Dec Accessed at: http://www.cms.gov/manuals/. [accessed 2023 Nov 09] [Context Link 1]
110. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12171 [Internet] Centers for Medicare and Medicaid Services. 2023 Aug 03 Accessed at: https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/. [accessed 2023 Oct 25] [Context Link 1]
111. Centers for Medicare and Medicaid Services. Medicare Program Integrity Manual. Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment Rev. 10365 [Internet] Centers for Medicare and Medicaid Services. 2020 Oct 02 Accessed at: https://www.cms.gov/regulations-and-guidance/regulations-and-guidance. [accessed 2023 Oct 20] [Context Link 1]
112. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: https://www.cms.gov/medicare-coverage-database/search.aspx? Updated 2023 [accessed 2023 Oct 20] [Context Link 1]

Footnotes

[A] Acute coronary syndrome describes the clinical spectrum from unstable angina to NSTEMI to STEMI.(5)(8) STEMI is characterized by symptoms of myocardial ischemia (eg, chest pain, dyspnea), ST-segment elevation on ECG, and an increased release of cardiac troponin from baseline.(5)(8) NSTEMI also includes the presence of symptoms of myocardial ischemia and increased cardiac troponin release from baseline, but without ST-segment elevation on ECG.(5)(8) [A in Context Link 1, 2]

[B] Acute myocardial infarction is classified into groups and subgroups by known or suspected etiology of ischemia and clinical setting.(5) Type 1 MI is due to atherosclerotic plaque rupture or erosion with subsequent thrombosis.(5) Type 2 MI is due to a mismatch between myocardial oxygen supply and demand; examples of decreased supply include coronary artery disease (nonruptured), coronary artery vasospasm, non-atherosclerotic coronary artery dissection, coronary embolism, microvascular dysfunction, and severe anemia, whereas examples of mismatch caused by increased demand include sustained tachyarrhythmia, severe bradyarrhythmia, respiratory failure, and hypertensive emergency.(5)(12) Type 3 MI is sudden cardiac death with symptoms or ECG changes suggestive of ischemia but death occurs before cardiac biomarkers are obtained or before cardiac biomarker values would be increased.(5) Type

4a MI occurs in the setting of PCI performed within the last 48 hours.(5) Type 4b is MI due to stent or scaffold thrombosis and type 4c is MI due to in-stent restenosis, both from previous PCI.(5) Type 5 is MI in the setting of coronary artery bypass grafting or other cardiac procedure performed in the previous 48 hours.(5) [B in Context Link 1, 2]

[C] The use of high-sensitivity troponin measurements are preferred due to more rapid and more accurate detection or exclusion of myocardial injury.(14) A report from a facility (eg, hospital) laboratory of an elevated troponin level, in general, indicates a level above the 99th percentile for the specific troponin assay used. This is important, as the definition and diagnosis of a myocardial infarction hinges on this being the case; that is to say, a troponin above the 99th percentile is a necessary component of the diagnosis. The precise value that represents the 99th percentile for cardiac troponin depends on the specific troponin assay used.(5)(8)(15) While universal agreement is absent, it is generally recommended that age-specific 99th percentile cutoffs not be used, but sex-specific cutoffs are recommended when available.(5) For contemporary (not high-sensitivity) cardiac troponin I and T assays, international standards as to what constitutes the 99th percentile for various assays have been set forth; see https://www.ifcc.org/media/477936/contemporary-cardiac-troponin-i-and-t-assay-analytical-characteristics-designated-by-manufacturer-v062019.pdf for more information. The same has been done for high-sensitivity assays; see https://www.ifcc.org/media/477937/high-sensitivity-cardiac-troponin-i-and-t-assay-analytical-characteristics-designated-by-manufacturer-v062019.pdf and for point of care assays, see https://www.ifcc.org/media/477940/point-of-care-cardiac-troponin-i-and-t-assay-analytical-characteristics-designated-by-manufacturer-v062019.pdf for more information. [C in Context Link 1, 2, 3]

[D] Troponin elevation alone does not indicate acute myocardial infarction, as patients can have elevated levels of cardiac troponin due to other causes of myocardial injury, either acutely or chronically.(5) Patients with elevated troponin levels without known baseline troponin levels should, upon subsequent measurements, have a troponin level rise or fall of more than 20% to confirm acuity of myocardial injury. Absent this variation, the acuity of the myocardial injury should be questioned (ie, patient may have chronic troponin elevation), as should the diagnosis of acute myocardial infarction.(5) [D in Context Link 1, 2, 3]

[E] Examples of acute nonischemic causes of myocardial injury include heart failure exacerbation, myocarditis, cardiac procedures, cardiac contusion, pulmonary embolism, and defibrillation shock.(5) Examples of chronic cardiac causes not due to acute ischemia would be cardiomyopathy, valvular disease, chronic heart failure, cardiac amyloidosis, or sarcoidosis.(5)(16) Patients with chronic kidney disease are particularly susceptible to chronic, nonischemic elevations of cardiac troponin levels.(5) In clinical situations where it is unknown if troponin release is due to ischemia or not, the dynamic nature of the troponin levels (rising or falling vs stable), presence or absence of associated symptoms, ECG changes, and imaging findings become even more important in diagnosing acute myocardial infarction.(5)(16) [E in Context Link 1, 2, 3]

[F] Serial cardiac troponin levels are generally measured 3 hours apart using standard assays, but dynamic changes in troponin can be detected in as little as 1 hour using high-sensitivity troponin assays, allowing for faster diagnosis of acute myocardial injury.(5)(8)(15) Because cardiac troponin levels are more sensitive and specific measures of cardiac injury than creatinine kinase myocardial band (CK-MB) measurements, CK-MB is not recommended for diagnosing acute coronary syndromes unless both conventional and high-sensitivity troponin measurements are unavailable.(3)(5)(8) [F in Context Link 1, 2, 3]

[G] Patients with suspected STEMI, right ventricular myocardial infarction, or posterior myocardial infarction may undergo invasive coronary angiography very soon after presentation, such that it will not be possible to await the results of sequential troponin testing (eg, looking for an acute rise or fall of 20%).(2)(5)(6)(7) Immediate invasive angiography is also recommended for patients with suspected myocardial infarction with hypotension, ventricular arrhythmia, acute heart failure, or myocardial ischemia refractory to medical treatment.(2)(5)(6)(7) In patients undergoing immediate invasive angiography, the diagnosis of myocardial infarction is informed by the results of the angiography along with the results of pre-catheterization troponin measurement (eg, drawn prior to or at the time of angiography). Other testing, such as cardiac imaging, can help confirm the diagnosis (eg, new wall motion abnormality).(2)(5)(6)(7) Emergent cardiac surgery may be necessary due to mechanical complications of myocardial infarction (eg, ventricular rupture, acute mitral regurgitation due to papillary muscle necrosis).(2)(7) [G in Context Link 1]

[H] Due to the likelihood of procedure-related myocardial injury, the troponin cutoff (ie, how high is high enough to indicate myocardial infarction) is higher than the cutoff for patients who have not recently undergone a cardiac procedure.(5) For example, if the cutoff for an elevated troponin is 0.03 ng/mL (mcg/L) (this number is only for the purposes of illustration and not to be taken as the actual cutoff for any given troponin assay) in patients who have not undergone a recent cardiac procedure, the cutoff would be 0.15 ng/mL (mcg/L) (5 times as high) for patients who have undergone PCI within the last 48 hours, and 0.30 ng/mL (mcg/L) (10 times as high) in patients who have undergone a cardiac procedure other than PCI (eg, transcatheter valve intervention, coronary artery bypass grafting) within the last 48 hours. [H in Context Link 1, 2, 3, 4]

[I] Admission to the ICU or coronary care unit is appropriate for MI patients with persistent chest pain, hemodynamic instability, new unstable or symptomatic arrhythmia or ECG finding, syncope or near syncope, pulmonary edema due to ischemia, new or worsening mitral regurgitation murmur, S3 or rales, new-onset bundle branch block, respiratory failure, or hemorrhagic complication (eg, from thrombolysis or antiplatelet therapy).(1)(18)(19)(20) See Intensive, Intermediate, and Telemetry Care Guidelines ISC for further information. [I in Context Link 1]

[J] Low-risk MI patients who have undergone successful PCI may be admitted directly to intermediate care or telemetry.(1)(11) See Intensive, Intermediate, and Telemetry Care Guidelines ISC for further information. [J in Context Link 1]

[K] See Clinical Indications for Admission to Inpatient Care in this guideline. [K in Context Link 1]

[L] Prothrombin time is necessary to monitor warfarin dosage. Partial thromboplastin time is needed for monitoring if patient is given unfractionated heparin. [L in Context Link 1]

[M] For STEMI, a specialty guideline recommends immediate (within 2 hours) primary PCI of culprit coronary lesions for patients whose symptom onset was within 24 hours of presentation, or for ongoing ischemia, heart failure, unstable rhythm disorder, or cardiogenic shock.(2) The same guideline recommends thrombolysis be used for STEMI only if PCI is not available within a 2-hour window, with subsequent transfer to a PCI-capable hospital for early angiography with intent to revascularize if needed.(2) [M in Context Link 1, 2]

[N] Hemodynamically stable patients with STEMI treated initially with PCI of the identified culprit artery may undergo PCI of non-culprit arteries either as part of the initial procedure, as a second procedure before discharge, or as a planned interval procedure following discharge.(2) [N in Context Link 1]

[O] Only patients who are hypoxic (eg, oxygen saturation less than 90%) should receive oxygen. [O in Context Link 1]

[P] Morphine may be indicated for continuing pain or dyspnea.(1) [P in Context Link 1]

[Q] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [Q in Context Link 1]

[R] Randomized studies have shown that influenza vaccination within 72 hours of hospitalization decreases the risk of subsequent major adverse cardiovascular events.(21) [R in Context Link 1, 2]

[S] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(77) [S in Context Link 1]

Definitions

Acute kidney injury (stage 2)

- Acute kidney injury (stage 2) with significant clinical findings, as indicated by **ALL** of the following(1)(2)(3):
  - Acute[A] kidney injury (stage 2), as indicated by **1 or more** of the following[B]:
    - Within past 7 days, rise in serum creatinine between 2 and 2.9 times its baseline value (increase of 3 times baseline would qualify as stage 3 acute kidney injury)
    - Decreased urine output,[C] as indicated by **ALL** of the following:
      - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
      - Urine output less than 0.5 mL/kg/hr for 12 hours
  - Significant clinical findings, as indicated by **1 or more** of the following[B]:
    - Hemodynamic instability
    - Worsening clinical status despite observation care (eg, rising creatinine or urine output remains less than 0.5 mL/kg/hr)
    - Altered mental status
    - Cardiac arrhythmias of immediate concern
    - Severe hypertension (SBP greater than 180 mm Hg or DBP greater than 120 mm Hg in adults or greater than the 95th percentile for age, gender, and height plus 30 mm Hg in pediatric patients)(4)(5)
    - Significant respiratory findings (eg, pulmonary edema) that are severe (eg, Hypoxemia) or persist despite observation care
    - Clinically significant electrolyte abnormality that is severe (eg, hyperkalemia with severe ECG findings)[D] or persists despite observation care
    - Clinically significant metabolic abnormality (eg, metabolic acidosis) that is severe or persists despite observation care

References

1. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney International. Supplement 2012;2(1):1-138. (Reaffirmed 2023 May)
2. Weisbord SD, Palevsky PM. Prevention and management of acute kidney injury. In: Yu AS, Chertow GM, Luyckx VA, Marsden PA, Skorecki K, Taal MW, editors. Brenner and Rector's The Kidney. 11th ed. Philadelphia, PA: Elsevier; 2020:940-977.e15.
3. Palevsky PM, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for acute kidney injury. American Journal of Kidney Diseases 2013;61(5):649-72. DOI: 10.1053/j.ajkd.2013.02.349.
4. Whelton PK, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Circulation 2018;138(17):e484-e594. DOI: 10.1161/CIR.0000000000000596. (Reaffirmed 2023 Jan)
5. Flynn JT, et al. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. Pediatrics 2017;140(3):e20171904. DOI: 10.1542/peds.2017-1904. (Reaffirmed 2022 Aug)

6. Mirvis DM, Goldberger AL. Electrocardiography. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:141-174.e3.
7. Mount DB. Fluid and electrolyte disturbances. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:338-356.

Footnotes

A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
B. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
C. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)
D. ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(6)(7)

Acute renal failure (stage 3 acute kidney injury)

- Acute[A] kidney injury (stage 3),[B] as indicated by **1 or more** of the following[C](1)(2)(3):
  - Within past 7 days, rise in serum creatinine to 3 times its baseline value or higher
  - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%) and serum creatinine is greater than 4 mg/dL (354 micromoles/L)
  - In child younger than 18 years, estimated glomerular filtration rate less than 35 mL/min/1.73m^2 (0.59 mL/sec/1.73m^2)
    eGFR - Pediatric Calculator and **1 or more** of the following:
    - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%)
    - Within past 48 hours, rise in serum creatinine greater than 0.3 mg/dL (26.5 micromoles/L)
  - Decreased urine output,[D] as indicated by **ALL** of the following:
    - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
    - Oligoanuria, as indicated by **1 or more** of the following:
      - Urine output less than 0.3 mL/kg/hour for 24 hours
      - Anuria (urine output less than 0.1 mL/kg/hour) for 12 hours
  - Initiation of renal replacement therapy (RRT) required(4)

References

1. Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney International. Supplement 2012;2(1):1-138. (Reaffirmed 2023 May)
2. Weisbord SD, Palevsky PM. Prevention and management of acute kidney injury. In: Yu AS, Chertow GM, Luyckx VA, Marsden PA, Skorecki K, Taal MW, editors. Brenner and Rector's The Kidney. 11th ed. Philadelphia, PA: Elsevier; 2020:940-977.e15.
3. Palevsky PM, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for acute kidney injury. American Journal of Kidney Diseases 2013;61(5):649-72. DOI: 10.1053/j.ajkd.2013.02.349.
4. Gaudry S, Palevsky PM, Dreyfuss D. Extracorporeal kidney-replacement therapy for acute kidney injury. New England Journal of Medicine 2022;386(10):964-975. DOI: 10.1056/NEJMra2104090.

Footnotes

A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
B. For purposes of these criteria, the term acute renal failure can be viewed as stage 3 acute kidney injury in the KDIGO classification system.(1) Of note, in the ICD-10 diagnostic coding system (main coding scheme used in labeling diagnoses), the relevant codes covering the clinical entity of significant acute kidney injury use the term acute kidney failure (eg, code N17.9).
C. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
D. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and

accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
  - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
  - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
  - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
  - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:117-125.e1.
3. Abraham G, Maher PJ. Delirium and dementia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1269-1280.e2.
4. Schor NF. Neurologic evaluation. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:3053-3063.e1.

Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
  - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
    - Resuscitated ventricular fibrillation or cardiac arrest
    - Ventricular escape rhythm
    - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)
    - Nonsustained ventricular tachycardia and **1 or more** of the following:
      - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
      - Acute myocarditis
  - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
    - Type II second-degree atrioventricular block
    - Third-degree atrioventricular block
    - New-onset left bundle branch block with suspected myocardial ischemia
  - Any heart rhythm and **1 or more** of the following(4)(5)(9)(10)(11):
    - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)
    - Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.
  - Heart rhythms of concern due to **1 or more** of the following:
    - Hypotension
    - Respiratory distress
    - Association with other significant symptoms (eg, syncope)(9)(10)(12)

References

1. Curtis AB, Tomaselli GF. Approach to the patient with cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1145-1162.e7.
2. Dalal AS, Van Hare GF. Disturbances of rate and rhythm of the heart. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:2436-2447.e1.
3. Joglar JA, et al. 2023 HRS Expert Consensus Statement on the management of arrhythmias during pregnancy. Heart Rhythm 2023;20(10):Online. DOI: 10.1016/j.hrthm.2023.05.017.
4. El-Chami M, Fernando L. Ventricular arrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1003-1014.
5. Al-Khatib SM, et al. 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Circulation 2018;138(13):e272-e391. DOI: 10.1161/CIR.0000000000000549. (Reaffirmed 2023 Jul)
6. Epstein AE, et al. 2012 ACCF/AHA/HRS focused update incorporated into the ACCF/AHA/HRS 2008 guidelines for device-based therapy of cardiac rhythm abnormalities: a report of the American College of Cardiology Foundation/American Heart Association

Task Force on Practice Guidelines and the Heart Rhythm Society. Circulation 2013;127(3):e283-e352. DOI: 10.1161/CIR.0b013e318276ce9b. (Reaffirmed 2022 Jun)

7. Miller JM, Ellenbogen KA. Therapy for cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1208-1244.
8. Park DS, Fishman GI. The cardiac conduction system. Circulation 2011;123(8):904-15. DOI: 10.1161/CIRCULATIONAHA.110.942284.
9. Joglar JA, et al. 2023 ACC/AHA/ACCP/HRS Guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. Journal of the American College of Cardiology 2023 Nov;online. DOI: 10.1016/j.jacc.2023.08.017. (Reaffirmed 2023 Nov 30)
10. Chun EB, McGorisk GM. Supraventricular tachyarrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:980-995.
11. Hoskins MH, De Lurgio DB. Pacemakers, defibrillators, and cardiac resynchronization devices in hospital medicine. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1025-34.
12. Westerman S, Manogue M, Hoskins MH. Bradycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:996-1002.

Care Task List

- Myocardial infarction care tasks to consider as soon as feasible may include:
  - Review results and care implications as appropriate:
    - Repeat cardiac biomarkers (eg, troponin)
    - Lipid profile
    - Telemetry findings
    - Repeat ECG
    - Other ordered investigations
  - Schedule, facilitate, or confirm not needed as appropriate:
    - Cardiology consultation
    - Angiography
    - Arterial sheath removal if PCI performed, as soon as feasible
    - Echocardiogram
    - Cardiac nuclear scan
    - Stress test
  - Advance toward planned diet.
  - Test saturation off oxygen and discontinue as appropriate.
  - Remove urinary catheter as soon as feasible.
  - Advance activity:
    - Out of bed
    - Ambulate, with telemetry if needed; confirm no symptoms are provoked.
  - Order as appropriate:
    - IV medication discontinuation, or adjustment and tapering instructions
    - Discontinuation of thrombolysis if used
    - Discontinuation of anticoagulation or transition to outpatient regimen
    - ACE inhibitor/ARB oral dosage, or document as not needed
    - Beta-blocker oral dosage, or document as not needed
    - Antiplatelet, oral nitrates, and statin outpatient regimen as appropriate
    - Conversion of all medications to oral route (or other outpatient regimen)
  - Confirm patient status:
    - No episodes of chest pain
    - Volume status stable on oral intake
    - Vital signs stable

Clinically active atrial fibrillation

- Clinically active atrial fibrillation refers to atrial fibrillation necessitating electrical cardioversion or treatment while hospitalized with parenteral (IV) medication to control heart rate (eg, digoxin, diltiazem, metoprolol) or to cardiovert (eg, amiodarone, procainamide).

Clinically active chronic obstructive pulmonary disease

- Clinically active chronic obstructive pulmonary disease (COPD) refers to COPD necessitating assisted ventilation (eg, BPAP, CPAP) or treatment while hospitalized with oral or parenteral (IV) corticosteroids.

Clinically active diabetes

- Clinically active diabetes refers to diabetes necessitating treatment with insulin while hospitalized.

Clinically active heart failure

- Clinically active heart failure refers to heart failure necessitating treatment with parenteral (IV) diuretics (eg, furosemide) during first 1 to 2 days of hospital care.

Dangerous arrhythmia

- Dangerous arrhythmia, as indicated by **1 or more** of the following(1)(2)(3):
  - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
    - Resuscitated ventricular fibrillation or cardiac arrest
    - Ventricular escape rhythm
    - Sustained (30 seconds or more of ventricular rhythm at greater than 100 beats per minute) ventricular tachycardia
    - Nonsustained ventricular tachycardia and **1 or more** of the following:
      - Acute myocarditis
      - Myocardial ischemia
  - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
    - Type II second-degree atrioventricular block
    - Third-degree atrioventricular block
    - New-onset left bundle branch block with suspected myocardial ischemia
  - Heart rhythms of concern due to **1 or more** of the following:
    - Hypotension
    - Respiratory distress
    - Association with other significant symptoms (eg, syncope)(9)(10)(11)

References

1. Curtis AB, Tomaselli GF. Approach to the patient with cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1145-1162.e7.
2. Dalal AS, Van Hare GF. Disturbances of rate and rhythm of the heart. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:2436-2447.e1.
3. Joglar JA, et al. 2023 HRS Expert Consensus Statement on the management of arrhythmias during pregnancy. Heart Rhythm 2023;20(10):Online. DOI: 10.1016/j.hrthm.2023.05.017.
4. El-Chami M, Fernando L. Ventricular arrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1003-1014.
5. Al-Khatib SM, et al. 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Circulation 2018;138(13):e272-e391. DOI: 10.1161/CIR.0000000000000549. (Reaffirmed 2023 Jul)
6. Epstein AE, et al. 2012 ACCF/AHA/HRS focused update incorporated into the ACCF/AHA/HRS 2008 guidelines for device-based therapy of cardiac rhythm abnormalities: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Heart Rhythm Society. Circulation 2013;127(3):e283-e352. DOI: 10.1161/CIR.0b013e318276ce9b. (Reaffirmed 2022 Jun)
7. Miller JM, Ellenbogen KA. Therapy for cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1208-1244.
8. Park DS, Fishman GI. The cardiac conduction system. Circulation 2011;123(8):904-15. DOI: 10.1161/CIRCULATIONAHA.110.942284.
9. Joglar JA, et al. 2023 ACC/AHA/ACCP/HRS Guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. Journal of the American College of Cardiology 2023 Nov;online. DOI: 10.1016/j.jacc.2023.08.017. (Reaffirmed 2023 Nov 30)
10. Chun EB, McGorisk GM. Supraventricular tachyarrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:980-995.
11. Westerman S, Manogue M, Hoskins MH. Bradycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:996-1002.

Dangerous arrhythmia absent

- Dangerous arrhythmia absent, as indicated by absence of **ALL** of the following(1)(2)(3)(4)(5)(6)(7)(8)(9)(10)(11):
  - Resuscitated ventricular fibrillation or cardiac arrest
  - Ventricular escape rhythm
  - Sustained (30 seconds or more of ventricular rhythm at greater than 100 beats per minute) ventricular tachycardia
  - Nonsustained ventricular tachycardia with myocarditis or ischemia
  - Type II second-degree atrioventricular block
  - Third-degree atrioventricular block
  - New-onset left bundle branch block with suspected myocardial ischemia

    - Heart rhythms of concern due to association with Hypotension, Respiratory distress, or other significant symptoms (eg, syncope)

References

1. Curtis AB, Tomaselli GF. Approach to the patient with cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1145-1162.e7.
2. Dalal AS, Van Hare GF. Disturbances of rate and rhythm of the heart. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:2436-2447.e1.
3. El-Chami M, Fernando L. Ventricular arrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:1003-1014.
4. Al-Khatib SM, et al. 2017 AHA/ACC/HRS guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Circulation 2018;138(13):e272-e391. DOI: 10.1161/CIR.0000000000000549. (Reaffirmed 2023 Jul)
5. Epstein AE, et al. 2012 ACCF/AHA/HRS focused update incorporated into the ACCF/AHA/HRS 2008 guidelines for device-based therapy of cardiac rhythm abnormalities: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines and the Heart Rhythm Society. Circulation 2013;127(3):e283-e352. DOI: 10.1161/CIR.0b013e318276ce9b. (Reaffirmed 2022 Jun)
6. Miller JM, Ellenbogen KA. Therapy for cardiac arrhythmias. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:1208-1244.
7. Park DS, Fishman GI. The cardiac conduction system. Circulation 2011;123(8):904-15. DOI: 10.1161/CIRCULATIONAHA.110.942284.
8. Joglar JA, et al. 2023 ACC/AHA/ACCP/HRS Guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. Journal of the American College of Cardiology 2023 Nov;online. DOI: 10.1016/j.jacc.2023.08.017. (Reaffirmed 2023 Nov 30)
9. Chun EB, McGorisk GM. Supraventricular tachyarrhythmias. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:980-995.
10. Westerman S, Manogue M, Hoskins MH. Bradycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:996-1002.
11. Joglar JA, et al. 2023 HRS Expert Consensus Statement on the management of arrhythmias during pregnancy. Heart Rhythm 2023;20(10):Online. DOI: 10.1016/j.hrthm.2023.05.017.

Extended Stay Risk Assessment

- Risk of extended length of stay is increased by presence of **1 or more** of the following(1):
    - Acute heart failure (eg, pulmonary edema)
    - Acute renal failure (stage 3 acute kidney injury)

References

1. Premier PINC AI™ Healthcare Database (PHD), 01/01/2021-12/31/2022. Premier, Inc.

Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
    - Vital sign abnormality not readily corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
        - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
        - Hypotension that persists despite appropriate treatment (eg, volume repletion, treatment of underlying cause)
        - Orthostatic hypotension that persists despite appropriate treatment (eg, volume repletion)
    - Hypotension that is severe, as indicated by **ALL** of the following:
        - Hypotension
        - Severe hypotension, as indicated by **1 or more** of the following:
            - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
            - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
            - Mean arterial pressure[A][D] less than 65 mm Hg
            - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

References

1. Puskarich MA, Jones AE. Shock. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:34-41.e1.
2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-114.

3. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.
4. Turner DA, Cheifetz IM. Shock. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:572-583.e1.
5. Nyhan A. Cardiology. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:145-188.e4.
6. Naidu SS, et al. SCAI SHOCK stage classification expert consensus update: a review and incorporation of validation studies: this statement was endorsed by the American College of Cardiology (ACC), American College of Emergency Physicians (ACEP), American Heart Association (AHA), European Society of Cardiology (ESC) Association for Acute Cardiovascular Care (ACVC), International Society for Heart and Lung Transplantation (ISHLT), Society of Critical Care Medicine (SCCM), and Society of Thoracic Surgeons (STS) in December 2021. Journal of the American College of Cardiology 2022;Online. DOI: 10.1016/j.jacc.2022.01.018.
7. Wardi G, Brice J, Correia M, Liu D, Self M, Tainter C. Demystifying lactate in the emergency department. Annals of Emergency Medicine 2020;75(2):287-298. DOI: 10.1016/j.annemergmed.2019.06.027.
8. Levitt DG, Levitt JE, Levitt MD. Quantitative assessment of blood lactate in shock: measure of hypoxia or beneficial energy source. BioMed Research International 2020;2020:2608318. DOI: 10.1155/2020/2608318.
9. Bakker J, Postelnicu R, Mukherjee V. Lactate: where are we now? Critical Care Clinics 2020;36(1):115-124. DOI: 10.1016/j.ccc.2019.08.009.
10. Kraut JA, Madias NE. Lactic acidosis. New England Journal of Medicine 2014;371(24):2309-2319. DOI: 10.1056/NEJMra1309483.
11. Olivieri L, Chasm R. Diabetic ketoacidosis in the pediatric emergency department. Emergency Medicine Clinics of North America 2013;31(3):755-773. DOI: 10.1016/j.emc.2013.05.004.
12. Maloney GE, Glauser JM. Diabetes mellitus and disorders of glucose homeostasis. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:1543-1558.e2.
13. Arnold TD, Miller M, van Wessem KP, Evans JA, Balogh ZJ. Base deficit from the first peripheral venous sample: a surrogate for arterial base deficit in the trauma bay. Journal of Trauma 2011;71(4):793-7; discussion 797. DOI: 10.1097/TA.0b013e31822ad694.
14. Malinoski DJ, Todd SR, Slone S, Mullins RJ, Schreiber MA. Correlation of central venous and arterial blood gas measurements in mechanically ventilated trauma patients. Archives of Surgery 2005;140(11):1122-5. DOI: 10.1001/archsurg.140.11.1122.
15. Zakrison T, McFarlan A, Wu YY, Keshet I, Nathens A. Venous and arterial base deficits: do these agree in occult shock and in the elderly? A Bland-Altman analysis. Journal of Trauma and Acute Care Surgery 2013;74(3):936-9. DOI: 10.1097/TA.0b013e318282747a.

Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)
C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated.(12)
D. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals 1/3 SBP + 2/3 DBP.(1)(4) A calculator is available at https://www.mdcalc.com/mean-arterial-pressure-map.

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
  - Hemodynamic abnormalities at baseline or acceptable for next level of care
  - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
    - Tachycardia absent
    - Hypotension absent
    - No evidence of inadequate perfusion (eg, no myocardial ischemia)
    - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

1. Puskarich MA, Jones AE. Shock. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:34-41.e1.

2. Lewis J, Patel B. Shock. In: Gershel JC, Rauch DA, editors. Caring for the Hospitalized Child: A Handbook of Inpatient Pediatrics. 3rd ed. Elk Grove Village, IL: American Academy of Pediatrics; 2023:107-114.
3. Ingbar DH, Thiele H. Cardiogenic shock and pulmonary edema. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2250-2257.
4. Brant EB, Seymour CW, Angus DC. Sepsis and septic shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2241-2249.
5. Singer M, et al. The Third International Consensus definitions for sepsis and septic shock (Sepsis-3). Journal of the American Medical Association 2016;315(8):801-810. DOI: 10.1001/jama.2016.0287.

Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
  - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
  - Low blood pressure, as indicated by **1 or more** of the following:
    - SBP less than 90 mm Hg in adult or child 10 years or older[A]
    - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
    - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
    - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
    - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
2. Massaro AF. Approach to the patient with shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2235-2241.
3. Horeczko T. Pediatric cardiac disorders. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2109-2131.e1.
4. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP. A calculator is available at https://www.mdcalc.com/mean-arterial-pressure-map.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
  - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
  - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
  - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
  - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
  - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
2. Massaro AF. Approach to the patient with shock. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2235-2241.
3. Horeczko T. Pediatric cardiac disorders. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2109-2131.e1.

4. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
  - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO_2) of less than 90% or arterial blood gas partial pressure of oxygen (PO_2) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
  - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO_2 greater than 89% or PO_2 greater than 59 mm Hg (7.9 kPa)[A]
  - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

References

1. Dezube R, Lechtzin N. Hypoxia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:669-675.
2. Jamali H, et al. Racial disparity in oxygen saturation measurements by pulse oximetry: evidence and implications. Annals of the American Thoracic Society 2022;19(12):1951-1964. DOI: 10.1513/AnnalsATS.202203-270CME.
3. Rojas-Camayo J, et al. Reference values for oxygen saturation from sea level to the highest human habitation in the Andes in acclimatised persons. Thorax 2018;73(8):776-778. DOI: 10.1136/thoraxjnl-2017-210598.

Footnotes

A. Pulse oximetry (SpO2) may underestimate arterial saturation (SaO2), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Myocardial Infarction Multiple Condition Management Guidelines

- For patients who also have:
  - Clinically active atrial fibrillation, see Myocardial Infarction with Clinically Active Atrial Fibrillation MCM
  - Clinically active chronic obstructive pulmonary disease, see Myocardial Infarction with Clinically Active Chronic Obstructive Pulmonary Disease MCM
  - Clinically active diabetes and Clinically active heart failure, see Myocardial Infarction with Clinically Active Diabetes and Clinically Active Heart Failure MCM
  - Clinically active diabetes, see Myocardial Infarction with Clinically Active Diabetes MCM
  - Clinically active heart failure, see Myocardial Infarction with Clinically Active Heart Failure MCM
  - Severe renal failure, see Myocardial Infarction with Severe Renal Failure MCM

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
  - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
  - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

References

1. Shibao C, Lipsitz LA, Biaggioni I, American Society of Hypertension Writing Group. Evaluation and treatment of orthostatic hypotension. Journal of the American Society of Hypertension 2013 Jul-Aug;7(4):317-324. DOI: 10.1016/j.jash.2013.04.006.
2. Dalal AS, Van Hare GF. Syncope. In: Kliegman RM, St. Geme JW, Blum NJ, Shah SS, Tasker RC, Wilson KM, editors. Nelson Textbook of Pediatrics. 21st ed. Philadelphia, PA: Elsevier; 2020:566-571.e1.
3. Fang JC, O'Gara PT. History and physical examination: an evidence-based approach. In: Libby P, Bonow RO, Mann DL, Tomaselli GF, Bhatt DL, Solomon SD, editors. Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine. 12th ed. Elsevier; 2022:123-140.

Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Readmission Risk Assessment

- Risk of readmission is increased by presence of **1 or more** of the following(1)(2)(3)(4)(5)(6)(7)(8):
  - Hospitalization (nonelective) in past 6 months(9)(10)(11)(12)
  - 2 or more emergency department visits in past 6 months(9)
  - No source of outpatient care other than emergency department (eg, no primary care provider)(12)(13)
  - Severe care transition barriers (eg, no caregiver, homeless)(10)(11)(14)
  - New-onset atrial fibrillation during hospitalization(15)
  - Age 80 years or older(16)
  - Symptomatic heart failure (past or present)(16)
  - Symptomatic COPD(17)
  - Anemia (hematocrit less than 30% (0.30))
  - Serum sodium less than 130 mEq/L (mmol/L)
  - Renal insufficiency (GFR less than 60 mL/min/1.73m^2 (1.00 mL/sec/1.73m^2))(10)(18) eGFR - Adult Calculator
  - Metastatic solid tumor (eg, lung cancer, breast cancer)
  - Advanced liver disease (eg, cirrhosis with portal hypertension, history of variceal bleed)

References

1. MarketScan Database, 2012-2013 (Copyright @2012-2013 Truven Health Analytics Inc. All Rights Reserved.); proprietary health insurance data sources (2013-2014); and Medicare 100% Standard Analytical File (2013).
2. Preyde M, Brassard K. Evidence-based risk factors for adverse health outcomes in older patients after discharge home and assessment tools: a systematic review. Journal of Evidence-Based Social Work 2011;8(5):445-68. DOI: 10.1080/15433714.2011.542330.
3. Billings J, Dixon J, Mijanovich T, Wennberg D. Case finding for patients at risk of readmission to hospital: development of algorithm to identify high risk patients. British Medical Journal 2006;333(7563):327. DOI: 10.1136/bmj.38870.657917.AE.
4. Krumholz HM, et al. An administrative claims measure suitable for profiling hospital performance based on 30-day all-cause readmission rates among patients with acute myocardial infarction. Circulation. Cardiovascular Quality and Outcomes 2011;4(2):243-252. DOI: 10.1161/CIRCOUTCOMES.110.957498.
5. Kociol RD, et al. International variation in and factors associated with hospital readmission after myocardial infarction. Journal of the American Medical Association 2012;307(1):66-74. DOI: 10.1001/jama.2011.1926.
6. Hasan O, et al. Hospital readmission in general medicine patients: a prediction model. Journal of General Internal Medicine 2010;25(3):211-9. DOI: 10.1007/s11606-009-1196-1.
7. Billings J, Blunt I, Steventon A, Georghiou T, Lewis G, Bardsley M. Development of a predictive model to identify inpatients at risk of re-admission within 30 days of discharge (PARR-30). BMJ Open 2012;2(4):e001667. DOI: 10.1136/bmjopen-2012-001667.
8. Khera R, et al. Comparison of readmission rates after acute myocardial infarction in 3 patient age groups (18 to 44, 45 to 64, and =65 Years) in the United States. American Journal of Cardiology 2017;120(10):1761-1767. DOI: 10.1016/j.amjcard.2017.07.081.
9. van Walraven C, et al. Derivation and validation of an index to predict early death or unplanned readmission after discharge from hospital to the community. Canadian Medical Association Journal 2010;182(6):551-7. DOI: 10.1503/cmaj.091117.
10. Jencks SF, Williams MV, Coleman EA. Rehospitalizations among patients in the Medicare fee-for-service program. New England Journal of Medicine 2009;360(14):1418-28. DOI: 10.1056/NEJMsa0803563.

11. Garcia-Perez L, Linertova R, Lorenzo-Riera A, Vazquez-Diaz JR, Duque-Gonzalez B, Sarria-Santamera A. Risk factors for hospital readmissions in elderly patients: a systematic review. Quarterly Journal of Medicine 2011;104(8):639-51. DOI: 10.1093/qjmed/hcr070.
12. Woz S, et al. Gender as risk factor for 30 days post-discharge hospital utilisation: a secondary data analysis. BMJ Open 2012;2(2):e000428. DOI: 10.1136/bmjopen-2011-000428.
13. Jack BW, et al. A reengineered hospital discharge program to decrease rehospitalization: a randomized trial. Annals of Internal Medicine 2009;150(3):178-87.
14. Arbaje AI, Wolff JL, Yu Q, Powe NR, Anderson GF, Boult C. Postdischarge environmental and socioeconomic factors and the likelihood of early hospital readmission among community-dwelling Medicare beneficiaries. Gerontologist 2008;48(4):495-504.
15. Kundu A, et al. Relation of atrial fibrillation in acute myocardial infarction to in-hospital complications and early hospital readmission. American Journal of Cardiology 2016;117(8):1213-1218. DOI: 10.1016/j.amjcard.2016.01.012.
16. Brown JR, Conley SM, Niles NW. Predicting readmission or death after acute ST-elevation myocardial infarction. Clinical Cardiology 2013;36(10):570-575. DOI: 10.1002/clc.22156.
17. Campo G, et al. Impact of COPD on long-term outcome after ST-segment elevation myocardial infarction receiving primary percutaneous coronary intervention. Chest 2013;144(3):750-7. DOI: 10.1378/chest.12-2313.
18. Silverstein MD, Qin H, Mercer SQ, Fong J, Haydar Z. Risk factors for 30-day hospital readmission in patients =65 years of age. Proceedings (Baylor University. Medical Center) 2008;21(4):363-72.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
  - Patient with **1 or more** of the following:
    - Dyspnea (difficulty breathing)
    - Tachypnea
    - Abnormal breathing pattern (eg, chest retractions)
    - Other evidence of difficulty breathing
  - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
    - Hypoxemia
    - Altered mental status
    - Other evidence of respiratory compromise (eg, respiratory acidosis)

References

1. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
2. Naureckas ET, Solway J. Disturbances of respiratory function. In: Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson JL, editors. Harrison's Principles of Internal Medicine. 21st ed. McGraw Hill Education; 2022:2133-2139.

Severe renal failure

- Severe renal failure refers to renal failure or insufficiency necessitating inpatient dialysis or treatment with phosphate-binding agent medication (eg, sevelamer), or medication to treat secondary hyperparathyroidism (eg, cinacalcet).

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
  - Housing insecurity, as indicated by **1 or more** of the following:
    - Individual or caregiver's current living situation is **1 or more** of the following(5):
      - Does not have own housing (eg, staying in a hotel, shelter, or with others)
      - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
      - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
    - Current housing has **1 or more** of the following:
      - Electrical appliances (eg, stove, refrigerator) not working or unavailable
      - Insufficient heating or cooling
      - Insufficient ventilation
      - Lead paint or pipes
      - Mold
      - Pests (eg, bugs) or rodents
      - Smoke detectors not working or unavailable
  - Food insecurity, as indicated by **1 or more** of the following(6):
    - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
    - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.

- Insufficient transportation, as indicated by **1 or more** of the following(7):
  - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
  - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
- Insufficient utilities, as indicated by **1 or more** of the following(8):
  - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
  - In the past year, electric, water, gas, or oil company threatened to shut off services.
- Personal safety risk, as indicated by **2 or more** of the following(6):
  - Individual is sometimes or frequently physically hurt by another person (including family member).
  - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
  - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
  - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
- Insufficient dependent care, as indicated by **1 or more** of the following:
  - In the past year, individual or caregiver was unable to work due to lack of dependent care.
  - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
  - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
  - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
  - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
  - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

References

1. Scanlon A, Reinisch C. Social determinants of health. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems. 12th ed. St. Louis, MO: Mosby; 2023:18-20.
2. Moen M, Storr C, German D, Friedmann E, Johantgen M. A review of tools to screen for social determinants of health in the United States: a practice brief. Population Health Management 2020;23(6):422-429. DOI: 10.1089/pop.2019.0158.
3. Daniel-Robinson L, Moore JE. Innovation and Opportunities to Address Social Determinants of Health in Medicaid Managed Care. [Internet] Institute for Medicaid Innovation. 2019 Jan Accessed at: https://medicaidinnovation.org/emerging-topics/. [accessed 2023 Oct 20]
4. Billioux A, Verlander K, Anthony S, Alley D. Standardized Screening for Health-Related Social Needs in Clinical Settings: the Accountable Health Communities Screening Tool. [Internet] National Academy of Sciences. 2017 May Accessed at: https://nam.edu/. [accessed 2023 Sep 18]
5. Sandel M, et al. Unstable housing and caregiver and child health in renter families. Pediatrics 2018;14(2):e20172199. DOI: 10.1542/peds.2017-2199.
6. Children's HealthWatch Survey. Screening Instrument [Internet] Children's HealthWatch. 2020 Sep Accessed at: https://childrenshealthwatch.org/. [accessed 2023 Sep 11]
7. PRAPARE®: Protocol for Responding to and Assessing Patient Assets, Risks, and Experiences Screening Tool. [Internet] Association of Asian Pacific Community Health Organizations (AAPCHO) and National Association of Community Health Centers (NACHC). 2016 Sep Accessed at: https://prapare.org/the-prapare-screening-tool/. [accessed 2023 Sep 21]
8. Cook JT, et al. A brief indicator of household energy security: associations with food security, child health, and child development in US infants and toddlers. Pediatrics 2008;122(4):e867-75. DOI: 10.1542/peds.2008-0286.

Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
  - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
  - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
  - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
  - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
  - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
  - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia[A][B] absent, as indicated by **1 or more** of the following(1)(2):
  - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
  - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
  - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
  - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
  - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
  - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

References

1. Southmayd GL. Tachycardia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:729-739.
2. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea,[A][B] as indicated by respiratory rate of **1 or more** of the following(2)(3):
  - Greater than 18 breaths per minute in adult or child 13 years of age or older[A]
  - Greater than 22 breaths per minute in child 6 to 12 years of age[A]
  - Greater than 25 breaths per minute in child 3 to 5 years of age[A]
  - Greater than 30 breaths per minute in child 1 or 2 years of age[A]
  - Greater than 40 breaths per minute in infant 6 to 11 months of age[A]
  - Greater than 45 breaths per minute in infant 3 to 5 months of age[A]
  - Greater than 60 breaths per minute in infant 1 or 2 months of age[A]

References

1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Schafer AI, editors. Goldman-Cecil Medicine. 26th ed. Elsevier; 2020:28-31.e2.

2. Braithwaite SA, Wessel AL. Dyspnea. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:194-201.e2.
3. Pediatric parameters and equipment. In: Kleinman K, McDaniel L, Molloy M, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 22nd ed. 202: Elsevier; 2021:frontpiece tables.

Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(1) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(1)

Codes

ICD-10 Diagnosis: I21.01, I21.02, I21.09, I21.11, I21.19, I21.21, I21.29, I21.3, I21.4, I21.9, I21.A1, I21.A9, I21.B, I22.0, I22.1, I22.2, I22.8, I22.9, I23.3 [Hide]

CPT®: 0659T, 92941

CPT copyright 2023 American Medical Association. All rights reserved.

MCG Health
Inpatient & Surgical Care 28th Edition
Copyright © 2024 MCG Health, LLC
All Rights Reserved

Last Update: 3/14/2024 11:10:43 AM
Build Number: 28.1.20240313235330.016020